

PMP® Exam Questions & Answers

1. Define Project Scope.

- a. It is the design of experiments that are used to complete the project work.
- b. It is the combination of the cost and the schedule which is required to complete the project work.
- c. It is the description of the required work that is necessary to complete the project. [Ans]
- d. It is the description of the required work and resources that are needed to complete the project.

2. What is a Program?

- a. A very large and complex project.
- b. A collection of projects which have common resources. [Ans]
- c. A collection of sub-projects having a common goal.
- d. A collection of subprojects having a common customer.

3. Which of the following comprises the Project Life Cycle?

- a. Phases [Ans]
- b. Milestones
- c. Estimates
- d. Activities

4. Being the project manager of the ABC Project, you have allowed subsequent project phase to begin even before the predecessor phase completes. Which of the following relates to this scenario?

- a. Crashing
- b. Fast-tracking [Ans]
- c. Risk management
- d. Tandem scheduling

5. The person or group providing the resources and support for the project, program or portfolio and is also responsible for enabling success is called the:

- a. Project Manager
- b. Senior Management
- c. Sponsor [Ans]
- d. Client

6. Which of the following cannot be the part of Group creativity techniques:

- a. Vendor Bid Analysis [Ans]
- b. Brainstorming
- c. Affinity Diagram
- d. Nominal Group Technique

7. Being the project manager of a thermal power plant project you need to manage the construction of an administrative unit building. Which of the following technique will you use to quickly estimate and decide the use of historic estimate and expert judgement?

- a. Analogous Estimating [Ans]
- b. Parametric Estimating
- c. Historical Estimating
- d. Expert Judgement

8. What is the acceptable range used for determining the realistic activity cost estimates?

- a. Level of Accuracy [Ans]
- b. Units of Measure
- c. Level of Precision
- d. Cost Management Plan

9. What is the SPI of a Software development Project where EV = \$6,000, PV = \$5,000, AC = \$4,000?

- a. 0.73
- b. 1.2 [Ans]
- c. 0.8
- d. 1

10. Cost baseline is the output of which of the following?

- a. Plan Cost Management
- b. Estimate Costs
- c. Determine Budget [Ans]
- d. Control costs

11. Suppose you are the project manager on a thermal power plant project. How will you manage your team members, if there is a difference of opinion over the design of turbine?

- a. Organize meetings to resolve the misunderstanding and agree on a common solution. [Ans]
- b. Defer the decision to procure turbine components as much as possible.
- c. Copy the design of turbine from a similar past project.
- d. Procure the turbine parts from the nearest located vendor.

12. You find that activity in your project is very risky. How will you estimate the duration of this activity when your team member has given the optimistic, most likely and pessimistic duration for the activity?

- a. Using variance analysis
- b. Using Sensitivity analysis
- c. Using PERT formula [Ans]
- d. Using Performance Reports

13. Which of the following is not valid according to McGregor's Theory X?

- a. Employees seek to be directed.
- b. Employees avoid work.
- c. Employees are self-centered.
- d. Employees are self-motivated. [Ans]

14. You are the project manager for a software development project. Your CPI drops to 0.93 when one of your key developers suddenly falls ill. To make up for your affected work, you have decided to outsource it to a vendor. What will be the most appropriate contract type in this case:

- a. Retainage
- b. FPEPA
- c. Cost Reimbursable Contract [Ans]
- d. Letter of Intent

15. Who among the following said, "People will be interested in their safety needs only after their physiological needs have been fulfilled.":

- a. Maslow [Ans]
- b. Ouchi
- c. McGregor
- d. Herzberg

16. WBS is an excellent and most effective tool that is used for tracking of:

- a. Project Resources
- b. Project Schedule
- c. Project Scope [Ans]
- d. Project Risks

17. The best project organization structure for a small but highly technical project will be:

- a. Matrix Organization
- b. Functional Organization [Ans]
- c. Mixed Organization
- d. Projectized Organization

18. Being a project manager is you need to manage a project where there will be a number of persons working together. If you want to enhance the ability of the team to work together and perform as a team which one of the following things you need to do to maximize the ability of the team?

- a. Work breakdown structure
- b. Staffing Plan
- c. Cohabitation
- d. Colocation [Ans]

19. What is the role of the change control board?

- a. Assessing the impact of the change on Project objectives. [Ans]
- b. Defining requirements for the customer.
- c. Doing performance appraisal for team members involved in implementing changes.
- d. Encouraging team members to raise more change requests.

20. Being the project manager on a STP project, you decide to respond to an identified risk by contracting out work. Which of the following will lead to the minimum risk of the buyer?

- a. CPFF (Cost Plus Fixed Fee)
- b. FFP (Firm Fixed Price) [Ans]
- c. CPPC (Cost Plus Percentage of Cost)
- d. T&M (Time & Material)

21. For which process the stakeholderâ€™s engagement assessment matrix is prepared as a part?

- a. Identify Stakeholders

- b. Manage Stakeholder Engagement
- c. Control Stakeholder Engagement
- d. Plan Stakeholder management [Ans]

22. Being a project manager you need to spend a significant amount of time in communications. Which of the following option contains the three basic elements of a communications mode?

- a. Verbal, Nonverbal and Written
- b. Sender, receiver, and message [Ans]
- c. Text, drawing, and picture
- d. Manager, worker, and instructions

23. Suppose you are assigned as the project manager of a 30 months project which is in the executing phase. What will be the best way to resolve a conflict among the technical experts in your team?

- a. Find the root cause of the disagreement. [Ans]
- b. Resolve the conflict in favor of the senior most technical expert.
- c. Hire an external consultant to enforce his decision.
- d. Remove this technical work from the scope of project work.

24. What should be your primary concern while adding resources to your project?

- a. Effect of adding resources on project end date.
- b. Effect of adding resources on S curve.
- c. Effect of adding resources on RAM.
- d. Effect of adding resources on Cost and Schedule. [Ans]

25. If John is working with his team to decide the degree to which cost estimates should be rounded up or down, then he is working to establish which of the following:

- a. Units of measure
- b. Level of Precision [Ans]
- c. Level of accuracy
- d. Control Thresholds

26. Being a Project Manager, how will you deal with a stakeholder who has high power and high interest?

- a. Keep Satisfied
- b. Keep Informed
- c. Manage Closely [Ans]
- d. Monitor

27. While estimating the cost of work in a very new technology, being a project manager which of the following will concern you the most?

- a. Lack of Quality Policy
- b. Lack of Communications management plan
- c. Internet connection not reliable
- d. Lack of Historical Information [Ans]

28. A road project was planned to be completed in 8 months. Calculate the SV when the earned value information at the end of six months is given as: BAC = \$8000 , AC = \$12000, % complete = 100%.

- a. \$4,000
- b. \$2,000 [Ans]
- c. \$8,000
- d. \$6,000

29. What will be incentive amount for a contractor, who is working on a cost plus incentive fee contract? The project's target cost is \$5,000 with a target fee of \$800, the minimum fee is \$500 & the maximum fee is \$1,000. Also, the buyer-seller share ratio is 80:20 with an actual cost of \$6000.

- a. (\$200) [Ans]
- b. \$200
- c. (\$600)
- d. \$600

30. Which of the following a resource histogram shows but a responsibility assignment matrix does not?

- a. Activities
- b. Interrelationship
- c. Time [Ans]
- d. Activities assigned to

31. Which of the following is used to display sensitivity analysis?

- a. Sensitivity Diagram
- b. Data Flow Diagram
- c. Decision Tree
- d. Tornado Diagram [Ans]

32. If SPI = 0.75 and CPI = 0.8, then which of the following reports is correct?

- a. SV = \$3750

- b. $CV = \$4000$
- c. $TCPI = 0.8$
- d. The project likely needs more money than planned to complete [Ans]

33. According to PMBOK, project management processes are organized into which of the following order of process groups?

- a. Initiating, designing, planning, executing, controlling, and closing
- b. Designing, developing, testing, and implementing
- c. Initiating, planning, executing, monitoring and controlling, and closing [Ans]
- d. Initiating, planning, executing, monitoring, and closing

34. Which of the following statements is true regarding negative float?

- a. The project manager should try to remove negative float by crashing or fast-tracking. [Ans]
- b. There cannot be any negative float in the project.
- c. The project manager should try to remove negative float by reducing scope.
- d. It is desirable to have the negative float in the project.

35. Which of the following represents a set of conditions that should be satisfied before deliverables are accepted?

- a. Deliverables List
- b. Test Plan
- c. Acceptance Criteria [Ans]
- d. Punch List

36. How will a Pareto Chart help in a project with a lot of identified action items?

- a. Understand the trend of deviations
- b. Prioritization [Ans]
- c. Complete all activities on time
- d. Identify root causes

37. The activity relationship between the start of cleaning and the start of digging is represented by which of the following?

- a. Start-to-Start [Ans]
- b. Start-to-Finish
- c. Finish-to-Finish
- d. Finish-to-Start

38. Schedule Variance is:

- a. This always affects the CV.
- b. This never affects the CV.
- c. A negative value means that the project will likely be delayed. [Ans]
- d. A positive value means that the project will likely be delayed.

39. You are currently working as a Project Manager in Realty Logistics. You have been given responsibility for moving Infrastructure of ABC Software LTD to a new location. For that, you need to identify the stakeholders of your project. Based on the Power/Interest Grid that you have created, which of the following types of stakeholders' expectation should be managed closely?

- a. High Power – Low Interest
- b. High Power – High Interest [Ans]
- c. Low Power – Low Interest
- d. Low Power – High Interest

40. What is the company-wide policy which mandates all project quality known as?

- a. ISO 9000
- b. Quality planning
- c. Quality control
- d. Quality policy [Ans]

41. Which one of the following describes the difference between a Standard and a Deviation accurately?

- a. Standards are optional but regulations are not.
- b. Standards are mandatory but regulations are not.
- c. Regulations are mandatory but standards may be seen as guidelines. [Ans]
- d. Regulations and Standards are essentially the same.

42. What will be the BAC of your project, if it has a CPI of 0.80 and an EAC of \$ 1265825?

- a. \$1,000,000.00
- b. \$1,012,660.00 [Ans]
- c. \$1,582,281.00
- d. None of the above

43. The BEST way of making an accurate forecasting of ETC is:

- a. Manual Forecasting of the cost of remaining work [Ans]
- b. $BAC - AC$
- c. $EAC - AC$

d. $(BAC\ EV) / CPI$

44. How will you assess a situation where six observations are within the UPPER CONTROL LIMIT and control line while the seventh observation is observed within UPPER CONTROL LIMIT and Control line?

- a. There are no defects.
- b. Project Manager should move to the next process as everything is ok.
- c. The process is out of control. [Ans]
- d. The process is in control.

45. Which of the following can be used for Trend Analysis?

- a. Control Charts
- b. Cause and Effect Diagram
- c. Run Chart [Ans]
- d. Pareto Chart

46. In which of theory, Douglas McGregor has defined two models of worker behavior?

- a. Theory X and Theory Y [Ans]
- b. Theory Z
- c. Contingency Theory
- d. Expectancy Theory

47. What are the three 'I's' of stakeholders?

- a. Interest, influence, and involvement
- b. Interest, influence, and impact [Ans]
- c. Interest, influence, and importance
- d. Importance, inquiry, and influence

48. Which of the following is the objective of conducting Bidder Conferences?

- a. To make sure that all questions from potential sellers are answered privately.
- b. To make sure that potential sellers are treated as per the type of questions asked.
- c. To make sure that all potential sellers are treated equally and have access to the same information. [Ans]
- d. To ensure that all parties get answers to their questions.

49. What is the Japanese method of modern quality management called, which relies on continuous small improvements involving everyone from the top management to the lowest level worker in the organization?

- a. Kanban
- b. Kaizen [Ans]
- c. Deming Cycle
- d. PDCA

50. Herzberg divided motivation factors into two classes: satisfiers and dissatisfiers. Which of the following are the examples of satisfiers then?

- a. The sense of personal achievement, work satisfaction. [Ans]
- b. Plush office space, performance-based salary raise.
- c. Work satisfaction, a fringe benefit.
- d. Vacation time, assignment of a personal staff assistant.

PMP Math Questions

For the first three questions, assume you are managing the following project:

You are the project manager of a project that is budgeted to cost \$1 million at completion. The project will last ten months and its budget is spread evenly across each month. It is currently 50% complete and has so far cost \$750K but only produced value of \$500K.

1. What is the schedule variance (SV) of the project?

- A. \$0
- B. \$200K
- C. \$500K
- D. \$750K

2. What is the cost performance index (CPI) of the project?

- A. 0.67
- B. 1
- C. 1.5
- D. 2

3. If the variance is not expected to continue, what is the expected estimate at completion (EAC) of the project?

- A. \$500K
- B. \$750K
- C. \$1 million
- D. \$1.25 million

4. You want to estimate the duration of an activity on your project. Your pessimistic estimate is 10 days, your most likely estimate is 5 days, and your optimistic estimate is 3 days. Using a triangular PERT distribution, what is your duration estimate?

- A. 4.6 days
- B. 5 days
- C. 6 days
- D. 9.3 days

5. You are the project manager for a medium-sized project. There are 14 team members, and everyone communicates with everyone else on the project team. How many communication channels are there?

- A. 91
- B. 105
- C. 182
- D. 210

6. You are negotiating an FPIF contract. The ceiling price is \$2 million, and the target price is \$1.8 million. The buyer's share ratio is 75% for overruns. What is the target profit for the seller if the target cost for the buyer is \$1.5 million?

- A. \$150K
- B. \$200K
- C. \$300K
- D. \$1.5 million

7. Activity X takes 4 days and is followed by Activity Y, which takes 3 days. The early start of Activity X is day 20, and the early start of Activity Y is day 27. What is the free float of Activity X?

- A. 0
- B. 3
- C. 7
- D. 10

8. A project is expected to result in \$2 million in five years. The current interest rate is 5%. What is the PV of the project?

- A. \$1,359,252
- B. \$1,567,398
- C. \$1,784,972
- D. \$2 million

9. You must select one and only one project to take on for your company. The net present value (NPV) for each project is as follows: Project A's NPV is \$10K, Project B's NPV is \$20K, and Project C's NPV is \$30K.

- A. Project A
- B. Project B
- C. Project C
- D. This problem cannot be solved without knowing the interest rate for each project.

10. A new state-of-the-art computer for your project costs \$10K. You expect for it to last for four years and to sell it for \$1K for parts. How much should you book in depreciation each year for the computer? Assume you are using straight-line depreciation.

- A. \$1250
- B. \$2000
- C. \$2250
- D. \$2500

PMP Math Answers

The toughest of your PMP math will relate to EVM, or earned value management. That's why the first three questions are on EVM. These questions weren't so tough, but you can visit [7 Example PMP Earned Value Questions](#) for more information. Get to know it really well!

1. **A. \$0** – Schedule Variance = Earned Value – Planned Value, or $SV = EV - PV$. The project's current value at 50% completion is stated in the problem as \$500K. To determine the Planned Value, use the project's original BAC times percentage complete: $\$1 \text{ million} * 50\% = \$500K$. $SV = EV - PV = \$500K - \$500K = \$0$. The project is on schedule.

2. **A. 0.67** – Cost Performance Index = $CPI = EV/AC$. You determined EV in the previous problem at \$500K. The problem tells you that the project has so far cost \$750K, which is the AC. $CPI = EV/AC = \$500K/\$750K = 0.67$. The project is over budget.

3. **D. \$1.25 million** – The variance is expected to go away, so use the formula $AC + BAC - EV = \$700K + \$1 \text{ million} - \$500K = \1.2 million .

4. **C. 6 days** – Your typical PERT estimate is shown by the formula $(P + 4L + O)/6$, but this problem wants you to use the triangular PERT estimate instead. $(P + L + O)/3 = (10 + 5 + 3)/3 = 18/3 = 6 \text{ days}$. Did you use the wrong formula because you were moving too fast? Remember to slow down!

5. **B. 105** – The formula for determining the number of communication channels on a project is $[n(n-1)]/2$, where n is the number of project team members. Don't forget to include the project manager! In this problem, $n=15$, but you may have accidentally said $n=14$ if you forgot the PM.

6. **C. \$300K** – This problem is a good example of (1) not just memorizing the formula but knowing what goes into it and (2) avoiding extraneous information. The problem sets you up to think that you'll be using this formula: $PTA = [(Ceiling - Target)/Buyer's \text{ Share Ratio}] + Target$. Actually, you just need to know that the Target Price is made up of both the Buyer's Target Cost and the Seller's Target Profit. $\$1.8 \text{ million} - \$1.5 \text{ million} = \$300K$ in target profit for the seller.

7. **B. 3** – Network diagram math questions can be tricky, because they are almost like logic questions rather than math. For this one, know that free float (which is float on an activity) = $ES \text{ of following activity} - ES \text{ of present activity} - \text{present activity duration}$. We are determining the free float for Activity X. ES of following is easy because it is stated as 27. The duration of Activity X is also stated as four days. You need to use the ES information (day 20) and the duration (4 days). So since $\text{free float} = ES \text{ of following activity} - ES \text{ of present activity} - \text{present activity duration} = 27 - 20 - 4 = 3$.

8. **B. \$1,567,398** – Get those PV & FV formulas on your brain dump. Again, this is one of those problems that may or may not surface on your exam. It did for me! (But depreciation

didn't.) $\text{Present Value} = \text{FV} / [(1 + r)^n]$, where r is the rate of return and n is the number of years = $\$2 \text{ million} / [(1 + .05)^5]$.

9. **C. Project C** – For project evaluation dealing with internal rate of return (IRR), return on investment (ROI), and net present value (NPV), simply pick the greatest value. Easy!

10. **C. \$2250** – $(\text{Asset Cost} - \text{Scrap Value}) / \text{Useful Life} = (\$10\text{K} - \$1\text{K}) / 4 = \2250 per year. It is possible, though not certain, that you will encounter a depreciation problem on your PMP exam, though I happened not to encounter one. Add the formulas to your brain dump anyway — a point is a point.